

Title

Event Management API

A RESTful API built with **FastAPI** for managing personal events. The application provides secure user authentication using JWT tokens and allows authenticated users to create, manage, update, and delete their own events.

Features

- User registration
- User authentication with JWT
- Password hashing using bcrypt
- Protected API endpoints
- Create, retrieve, update and delete personal events
- Filter events by date or location
- User authorization (users can only access their own events)

Technologies Used

Python, FastAPI, SQLAlchemy, PostgreSQL, Pydantic, python-jose (JWT), Passlib (bcrypt)

Project Structure

```
main.py
database.py
security.py
models/: user.py, event.py
routers/: users.py, events.py
schemas/: user.py, event.py
```

Installation

1. Clone the repository.
2. Create a virtual environment.
3. Install dependencies:
`pip install fastapi uvicorn sqlalchemy psycopg2-binary python-jose passlib[bcrypt] python-multipart email-validator`
4. Configure DATABASE_URL in database.py.
5. Start the server:
`uvicorn main:app --reload`

API Documentation

<http://127.0.0.1:8000/docs>

Authentication

Register, log in to obtain a JWT token, then authorize using Bearer <your_token>.

Endpoints

Authentication

```
POST /register
POST /login
GET /me
```

Events

```
POST /events
GET /events
GET /events/{id}
PUT /events/{id}
DELETE /events/{id}
GET /events/filter
```

Security

Passwords are hashed with bcrypt. JWT protects private endpoints. Each user can access only

their own events.

Database

Entities: User and Event. Each event belongs to one user through a foreign key relationship.

License

This project is intended for educational purposes.